

Vaishnav Mukund Mankar

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SUMMARY

Computer Engineering student focused on building scalable AI and distributed systems. Experienced in Python, Rust, and cloud-native development with projects in NLP, LLM pipelines, and real-time applications, with strong foundations in system design and data structures.

EDUCATION

Pune Institute of Computer Technology (PICT)
B.E. Computer Engineering; CGPA: 9.2

Pune, India
2023 – 2027

EXPERIENCE

AI Developer Intern

Aug 2025 – Feb 2026

Across The Globe

Remote

- Engineered the backend infrastructure for an AI-driven video generation pipeline, delivering personalized multi-lingual content at scale.
- Optimized AWS Lambda execution efficiency, reducing processing time by 30% (from 20m to 14m) and significantly cutting compute costs.
- Architected internal automation workflows, including a Markdown-to-PDF engine and a document-grounded RAG system for instant knowledge retrieval.

PROJECTS

Real-Time AI Translation Chat Application

- Built a chat platform that translates medical conversations instantly between different languages for doctors and patients.
- Tech Stack:** Python, TypeScript, WebSockets, Whisper, and Google Gemini AI.
- Impact:** Achieved ultra-fast translation (under 200ms), ensuring seamless real-time communication.

Cloud-Native Developer Utility Suite

- Developed a suite of high-performance developer utilities including Markdown-to-PDF and LaTeX-to-PDF converters.
- Tech Stack:** SolidJS, TypeScript, and Serverless Functions.
- Impact:** Streamlined documentation workflows, achieving near-instant conversion and seamless user interaction.

RAG-Powered Educational AI Agent

- Created an AI tutor that helps students by using only verified textbook information to ensure 100% accurate answers.
- Tech Stack:** Python, LangChain, Vector Databases, and Large Language Models (LLMs).
- Impact:** Prevented AI "hallucinations" by grounding responses in official curriculum data.

AI-Powered Smart Photo Gallery

- Engineered a private "Google Photos" alternative that allows users to search for images using natural, descriptive sentences.
- Tech Stack:** React, FastAPI, PostgreSQL, Google Gemini, and Docker.
- Impact:** Enabled instant search and automatic image tagging while keeping 100% of user data on their own server.

ACHIEVEMENTS

IEEE Publication: Published "GNN-Based Friend Recommendation: A Graph Embedding Approach for Social Networks" in an IEEE International Conference.

TECHNICAL SKILLS

Languages: Python, Rust, SQL, TypeScript, JavaScript

Frameworks: FastAPI, React, Next.js, LangChain

Cloud & Tools: AWS/Azure, Git, Linux, PostgreSQL, MongoDB

Concepts: Data Structures & Algorithms, Distributed Systems, System Design, NLP, RAG